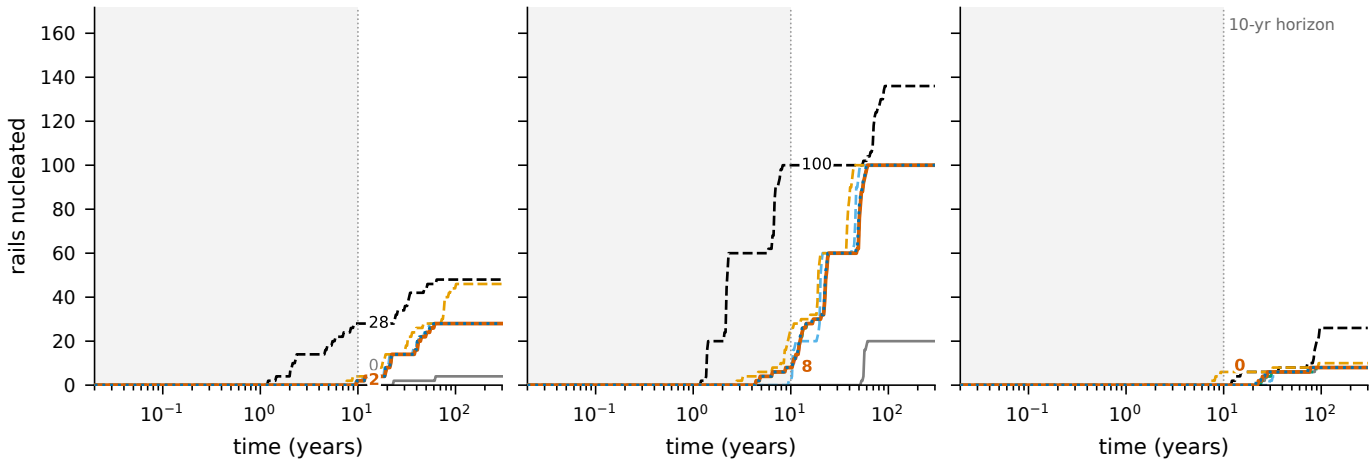


D1 (IO, 2 W)

D2 (SRAM, 4 W)

D3 (logic, 20 W)

10-yr horizon



— die-level sign-off (standalone thermal)

- - - uniform die temperature

- - - no wire self-heating

— full 3-D physics (FDM truth)

- - - rule sign-off (uniform 105 °C, const. σ_T)- - - σ_T frozen at 2-D temperature

- - - no thermomigration

..... StackEM (SKN + JCX)